

# ARUNKUMAR S

Chennai, Tamil Nadu, India

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## Education

**Indian Institute of Information Technology Design & Manufacturing, Kancheepuram**  
*B.Tech. Smart Manufacturing*

Aug 2023 – Present  
**CGPA: 7.3/10**

**Narayana Junior College, Nellore**  
*Higher Secondary Certificate (Class XII)*

Sept 2021 – May 2023  
**90%**

**Narayana Higher Secondary School, Nellore**  
*Secondary School Certificate (Class X)*

June 2020 – Apr 2021  
**99%**

## Technical Skills

**Languages:** Python, C, HTML, SQL

**Simulation:** ANSYS, LS-DYNA, HyperMesh

**CAD:** SolidWorks, CATIA, AutoCAD, Fusion 360

**Engineering:** GD&T, FEA, DFM/DFA, Prototyping

## Experience

**Eaura Pvt Ltd (Government-Approved Startup) | Founder / CEO**

**June 2025 – Present**

- Founded a government-approved robotics & automation startup; defined the technical vision for an AI-powered voice assistant supporting independent living for senior citizens.
- Leading end-to-end MVP development – system architecture, reminders/scheduling, emergency-support features – and hardware-software integration across a cross-functional team.

**SRJ Energy Technologies (IITM Research Park) | Research Intern**

**April 2025 – Present**

- Designed and validated high-capacity cylindrical battery cells and composite battery enclosures for EV applications using SolidWorks and ANSYS, improving strength-to-weight ratio and impact resistance.
- Performed thermal/structural optimization and developed validation protocols, enhancing safety, durability, and heat dissipation of enclosure designs.

**Neyveli Lignite Corporation India Limited | Project Research Fellow**

**May 2025 – Present**

- Designed a lignite separation and conveyor handling system in SolidWorks, reducing operational downtime by 20% through improved workflow and equipment layout.
- Integrated magnetic separation and vibrating screening mechanisms, improving stone-segregation accuracy and plant safety while reducing manual sorting.

## Projects

**Compact Agricultural E-Transportation Vehicle | Project Team Lead**

**Jan 2026 – April 2026**

- Led development of a farm-based electric transport vehicle for banana logistics, overseeing vehicle dynamics, suspension, steering, braking, and chassis integration.

**Mobile Banana Processing Unit | Student, Pratis, Mumbai**

**Dec 2024 – Mar 2025**

- Designed a SolidWorks-based processing unit for banana de-handing, cleaning, and packaging with improved portability and shelf-life performance.

## Research Publications

**Data-Driven Mechanical Characterization of Re-Entrant Auxetic Honeycomb Structures Using Polynomial Ridge Regression**

*ASME IMECE-INDIA 2026 (Accepted)*

**AI-Assisted Rock Detection & Mechanical Diversion System for Lignite Mining Conveyor Operations**

*ICoRD'27 (Under Review)*

**Low Velocity Impact Response of Advanced Composites**

*Journal Review Paper (Under Review)*

## Leadership / Position of Responsibility

- Department Lead, Placement Cell
- Team Lead, Smart India Hackathon
- Department Volleyball Team Captain

## Achievements & Certifications

- GATE 2026 – Production & Industrial Engineering (PI): Score 384, AIR 710
- Certified SolidWorks Associate & Professional – Dassault Systèmes
- Foundations of Project Management – Google
- Product Management – GeeksforGeeks